

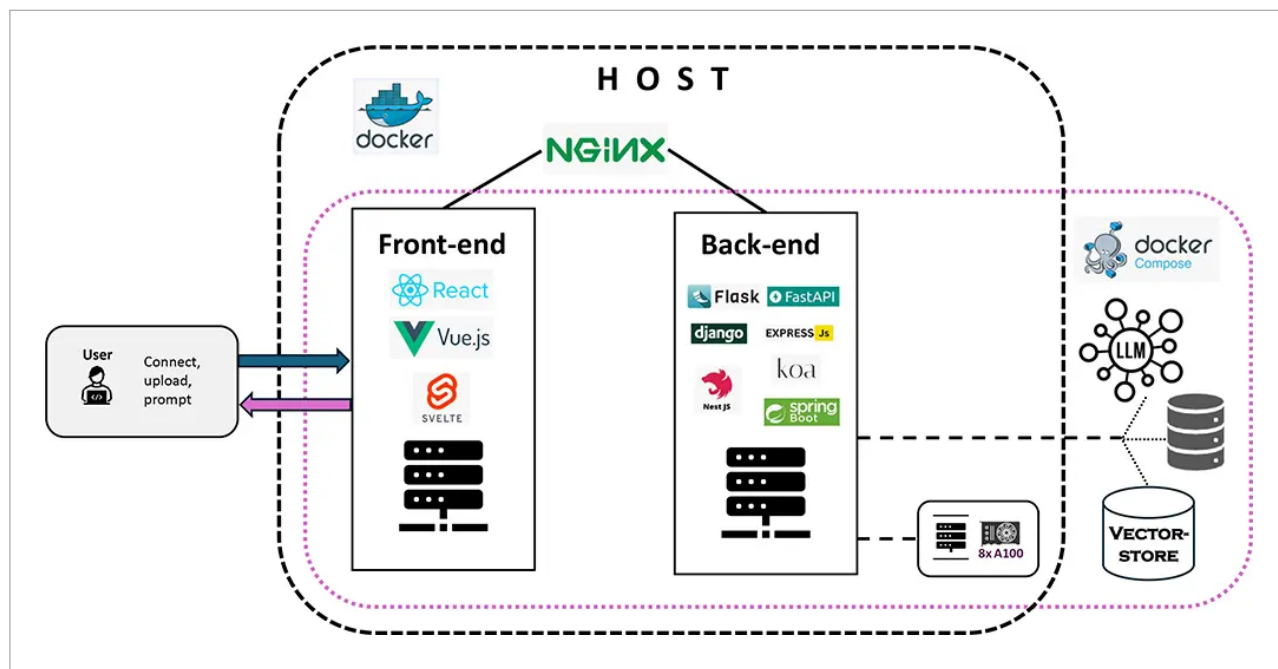


Figure 2: Docker-hosted AI/ML architecture

AI deployment — providing portability, consistency, and operational simplicity across hardware ecosystems.

Containerised AI enables developers and engineers to encapsulate models, runtime environments, and inference logic into isolated units that deploy with ease on edge devices — from industrial sensors and autonomous drones to hospital diagnostic equipment. This model obviates the historical friction between development and deployment, and harmonises with current needs for privacy, low latency, and real-time decision-making.

For production-ready deployments or edge scenarios involving caching and persistent storage, Docker Compose can orchestrate a lightweight AI stack as shown below:

```
version: '3.8'
services:
  ai-model:
    build: .
    ports:
      - "5000:5000"
    depends_on:
      - redis
      - mongodb
  redis:
    image: redis:alpine
    ports:
      - "6379:6379"
  mongodb:
```

```
image: mongo
ports:
  - "27017:27017"
```

Strategic advantages of Docker in edge AI

Portability across architectures: Docker containers encapsulate differences in underlying infrastructure. Developers can develop once on a development workstation and deploy throughout x86 or ARM-based systems — such as NVIDIA Jetson, Raspberry Pi, or cloud edge nodes — without environment-specific tuning.

Isolation and conflict resolution: Malicious coexistence of multiple ML services on edge platforms normally leads to conflicts of dependencies. Docker isolates services, ruling out package or runtime conflicts and making the system more reliable.

Maximised resource utilisation: Unlike virtual machines, Docker containers consume much less resources, which is critical in memory-constrained or battery-operated edge devices.

OTA (over-the-air) updates and rollbacks: Using Docker, AI models and business logic can be remotely updated through Docker registries or locked-down pipelines. This makes MLOps easier and improves maintainability of distributed AI systems.

Security and compliance: Containers provide process-level separation and support speedy patching of vulnerabilities. Also, Docker images can be signed and scanned to maintain integrity,